BIOLOGY REVISION NOTES

2 NOVEMBER TEST | CELL FOUNDATIONS

Learning focus: Know the structures before explaining specialised cells.

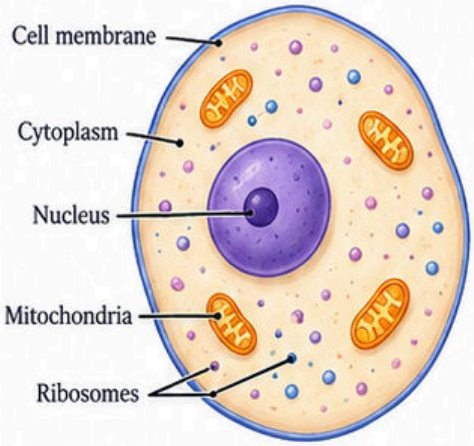
Based on school materials supplied 17 October.

1 KEY CELL STRUCTURES AND FUNCTIONS

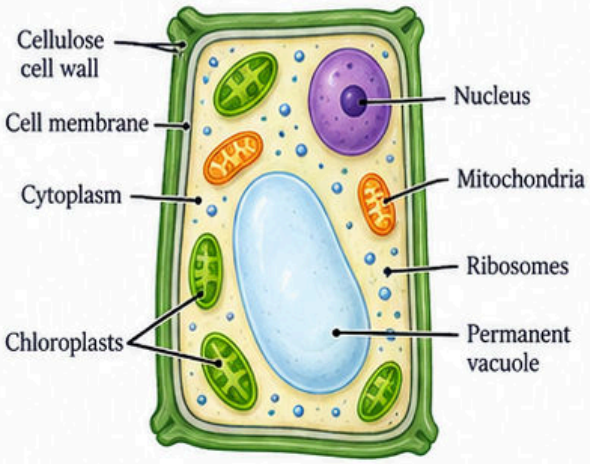
Structure	Function
Nucleus	contains genetic material and controls cell activities.
Cytoplasm	where most chemical reactions take place.
Cell membrane	controls movement of substances into and out of the cell.
Mitochondria	site of aerobic respiration, releasing energy for cell processes.
Ribosomes	site of protein synthesis.
Cellulose cell wall	strengthens and supports plant cells.
Chloroplasts	contain chlorophyll to absorb light for photosynthesis.
Permanent vacuole	contains cell sap; helps maintain turgor and support.

2 TYPICAL ANIMAL, PLANT AND BACTERIAL CELLS

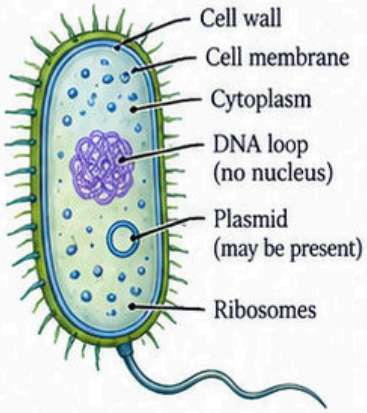
Animal cell (eukaryotic)



Plant cell (eukaryotic)



Bacterium (prokaryotic)



Bacteria have no mitochondria.

3 KEY COMPARISON

Plant and animal cells are **eukaryotic**: their DNA is enclosed in a nucleus.
Bacteria are **prokaryotic.**

4 MEMORY TIP

'Job first, structure second':

- protein production = **ribosomes**
- aerobic respiration = **mitochondria**
- photosynthesis = **chloroplasts.**

5 EXCEPTIONS TO REMEMBER

- Mature mammalian red blood cells lack a nucleus.
- Onion bulb epidermal cells and root hair cells usually lack chloroplasts.
- Not every plant cell photosynthesises.

6 GCSE-STYLE CHECK (2 marks)

Give two structures found in a typical plant cell but not in a typical animal cell.

Full-mark answer:
Cellulose cell wall [1]; **chloroplasts** [1].
A permanent vacuole is another acceptable example.

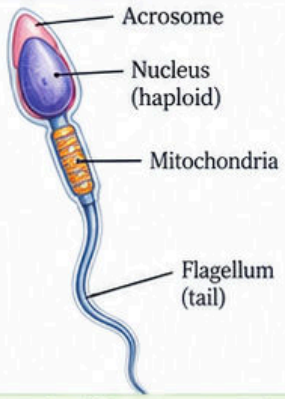
Answer tip: If asked for two, write two distinct structures. 'Cell wall' and 'cell membrane' are different.

BIOLOGY REVISION NOTES

SPECIALISED ANIMAL CELLS.

A **specialised cell** has structures that help it perform a particular function.
Differentiation is the process by which a cell becomes specialised.

1 SPERM

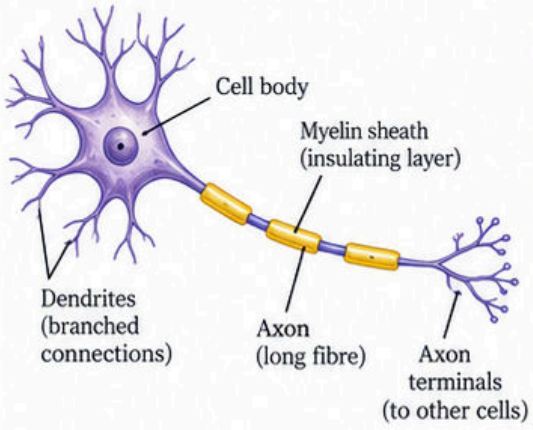


Function: fertilise an egg and deliver male genetic material.

- Flagellum propels it.
- Many mitochondria release energy for movement.
- Acrosome enzymes digest the egg's outer layers.
- Haploid nucleus has 23 chromosomes in humans.

Memory: Tail moves, mitochondria power, acrosome enters, nucleus delivers.

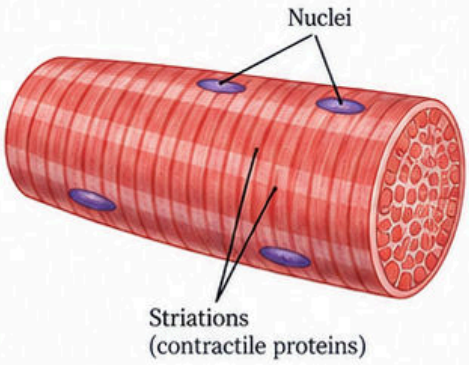
2 MOTOR NEURONE



Function: carries electrical impulses.

- Long axon carries impulses over long distances.
- Branched connections link to other cells.
- Myelin insulates the axon and increases transmission speed.

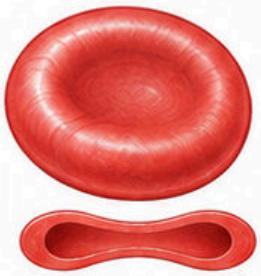
3 SKELETAL MUSCLE FIBRE



Function: contractile proteins slide past each other to shorten the cell; many mitochondria release energy for contraction.

- Skeletal muscle moves bones.
- Smooth muscle contracts in organ walls, such as the gut; it does not pull directly on bones.

4 RED BLOOD CELL

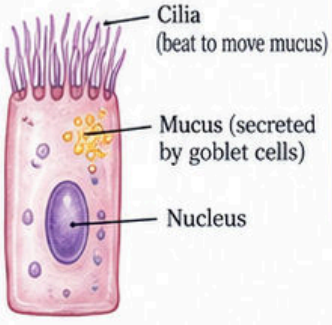


Top view (biconcave disc) Side view (shows biconcave shape)

Function: haemoglobin binds oxygen.

- Biconcave shape gives a large surface area relative to volume and a short diffusion distance.
- No nucleus leaves more space for haemoglobin.
- Flexible shape allows passage through narrow capillaries.

5 CILIATED EPITHELIAL CELL



Function: cilia beat to move mucus containing trapped particles and pathogens towards the throat.

- Mucus is secreted mainly by goblet cells.
- Cilia move it.

FULL-MARK METHOD

Structure → effect → function.

Do not just list structures.

GCSE-STYLE QUESTION (4 marks)

Explain **two** adaptations that help a sperm cell reach an egg. **[4 marks]**

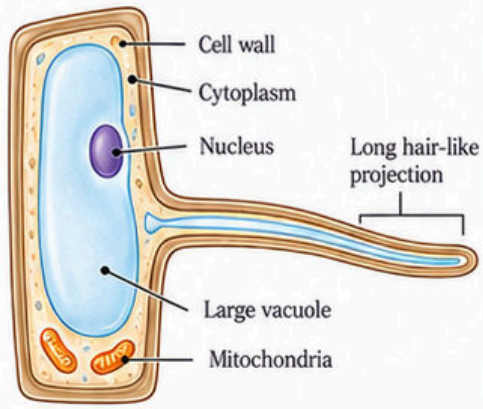
Model answer:
Its flagellum **[1]** propels the cell towards the egg **[1]**.
It has many mitochondria **[1]**, which release energy by respiration for movement **[1]**.

Common loss: 'Mitochondria make energy' is imprecise. Write 'release energy by respiration'. Acrosome enzymes help entry into the egg, rather than swimming towards it.

BIOLOGY REVISION NOTES

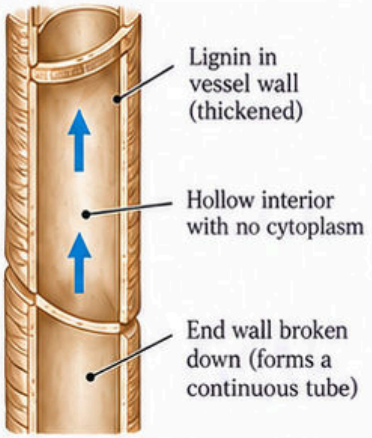
SPECIALISED PLANT CELLS.

1 ROOT HAIR CELL



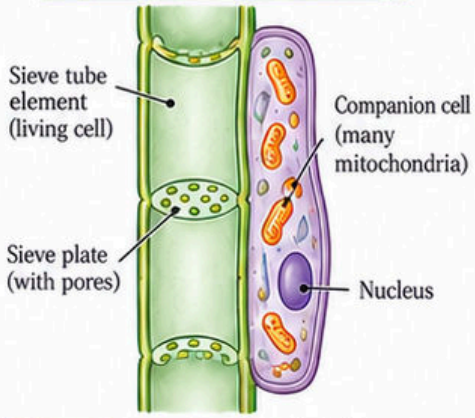
A root hair cell absorbs water and mineral ions from soil. A long projection gives a large surface area for absorption. Water enters by osmosis. Mineral ions can enter by active transport against a concentration gradient; mitochondria release the energy needed. Usually no chloroplasts because it is underground and receives little light.

2 XYLEM VESSEL SECTION



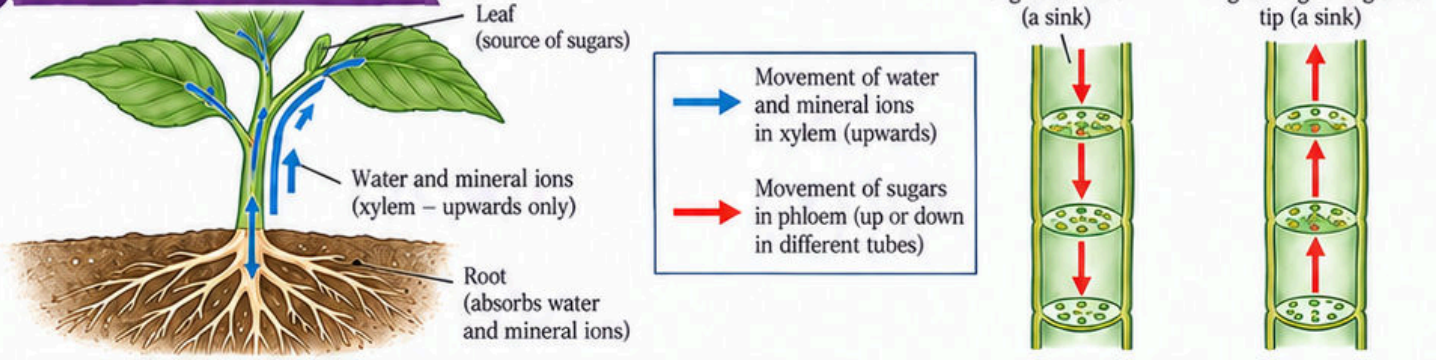
Xylem transports water and dissolved mineral ions upwards from roots; also supports the plant. Mature vessel elements are dead and hollow, with no cytoplasm obstructing flow. End walls break down, forming continuous tubes. Lignin strengthens and waterproofs vessel walls and helps prevent collapse.

3 PHLOEM SIEVE TUBE AND COMPANION CELL



Phloem transports dissolved sugars from sources, such as leaves, to sinks, such as growing or storage tissues. This is translocation. Transport may occur upwards or downwards in different sieve tubes. Living sieve tube elements have perforated sieve plates for flow and few internal structures. Companion cells contain many mitochondria, releasing energy for active loading of sugars.

4 TRANSPORT IN A PLANT



5 MEMORY COMPARISON

	Xylem	Phloem
What is transported?	Water and mineral ions	Dissolved sugars
Direction of transport	Mainly upwards (from roots to leaves)	From sources to sinks (can be upwards or downwards in different sieve tubes)
Type of cells	Dead, hollow vessel elements	Living sieve tube elements and companion cells
Key structural features	No cytoplasm (hollow tubes), lignin in walls	Perforated sieve plates; companion cells with many mitochondria

6 GCSE-STYLE QUESTION (4 MARKS)

Explain **two** ways xylem vessels are adapted for transporting water. [4 marks]

Model answer:

They form hollow tubes with no cytoplasm [1], so there is little obstruction to water flow [1]. Their walls contain lignin [1], which strengthens the vessels and helps prevent collapse [1].

7 SECOND CHECK (2 MARKS)

Explain why a root hair cell has a long projection. [2 marks]

Model answer:

It increases surface area [1], allowing greater absorption of water and mineral ions [1].

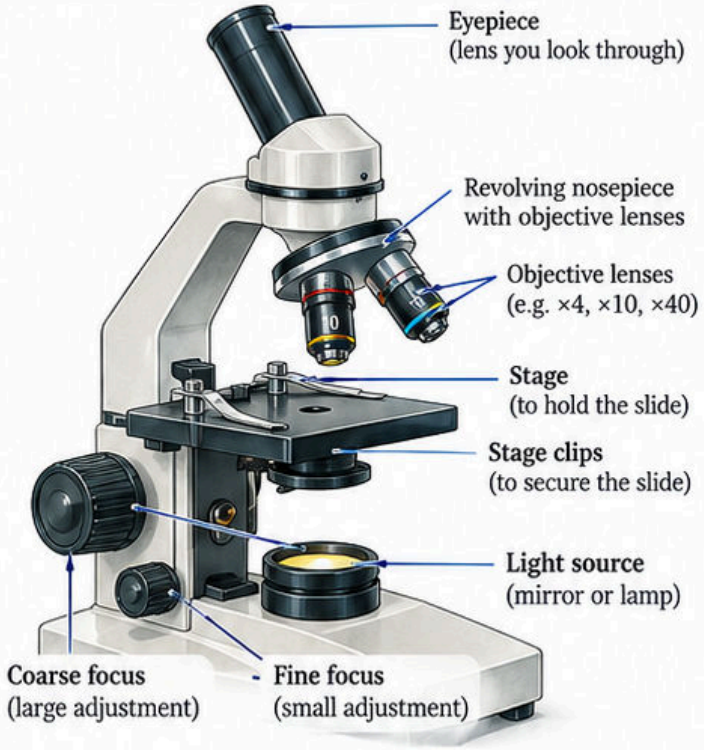
Answer tip: 'It absorbs more' alone misses the named structural feature. Keep water and mineral ions distinct: osmosis is water movement.

BIOLOGY REVISION NOTES

REQUIRED PRACTICAL 1 | LIGHT MICROSCOPY

Aim: Prepare a plant-cell slide, observe cells and produce a labelled drawing with a magnification scale. Also recognise animal-cell images.

THE LIGHT MICROSCOPE



STAIN COMPARISON

Iodine (for onion cells)



- Used for plant (onion) cells.
- Improves contrast so structures are easier to distinguish.

Methylene blue (*for animal cheek cells*)



- Commonly used for animal cheek cells.
- Improves contrast so structures are easier to distinguish.

An onion bulb cell has no chloroplasts because the bulb grows underground and does not photosynthesise.

SAFETY



- Handle glass and needles carefully.
- Stains may irritate and stain skin.
- Use eye protection as instructed.
- Do not force the objective into the slide.

GCSE-STYLE QUESTION (4 MARKS)

Describe how to prepare an onion epidermis slide. [4 marks]

Model answer (suggested marking points):

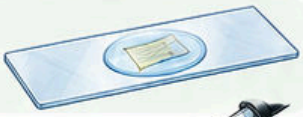
1. Peel a thin layer with forceps. [1]
2. Place it flat on a slide in water. [1]
3. Add iodine stain. [1]
4. Lower a coverslip at an angle to reduce air bubbles. [1]

METHOD WITH REASONS

1. Use forceps to peel a thin, single layer of onion epidermis: **a thin specimen lets light pass through and reduces overlapping cells.**



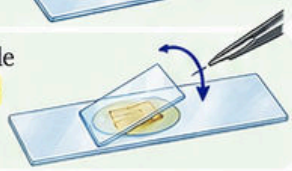
2. Place it flat in a drop of water on a clean slide.



3. Add iodine solution: staining improves contrast so structures are easier to distinguish.



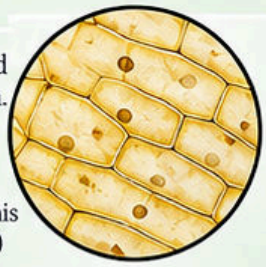
4. Lower a coverslip gently at an angle using a mounted needle: reduces trapped air bubbles.



5. Place the slide on the stage and secure with clips.
6. Select the lowest-power objective: larger field of view makes the specimen easier to locate.



7. Looking from the side, position the lens safely near the slide. Look through the eyepiece and focus while moving lens and slide apart. Use coarse focus, then fine focus.
8. Centre the specimen before increasing magnification. At high power, use fine focus and adjust illumination.
9. Draw what is visible; label observed structures and record magnification.



Onion epidermis cells (×400)

TROUBLESHOOT

No image?	Check light, specimen position and lowest objective.
Dark circles?	They may be air bubbles, not cells.



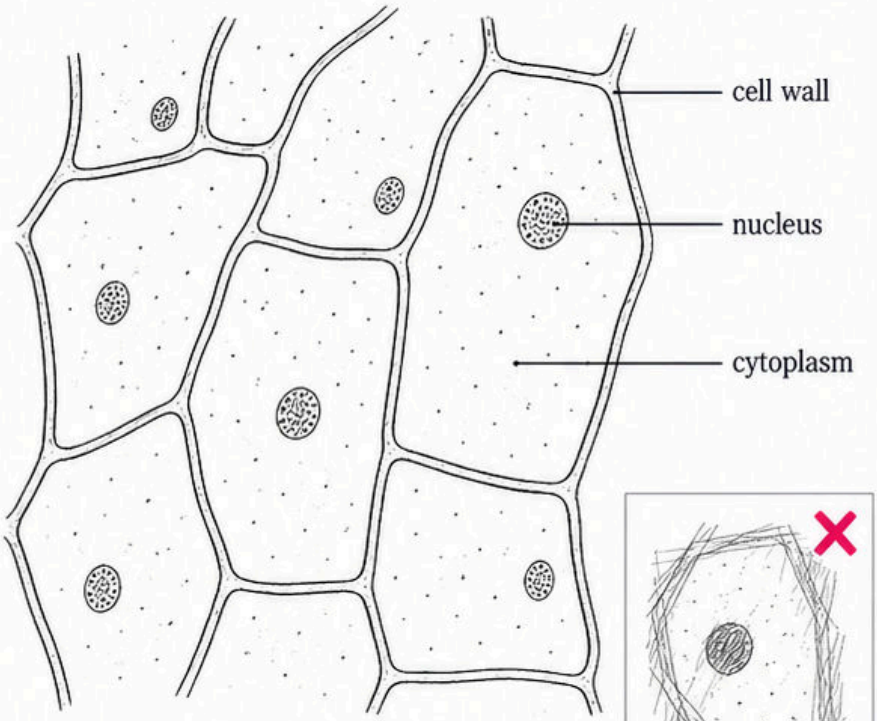
Air bubbles (not cells)

BIOLOGY REVISION NOTES

BIOLOGICAL DRAWING AND IMAGE QUALITY

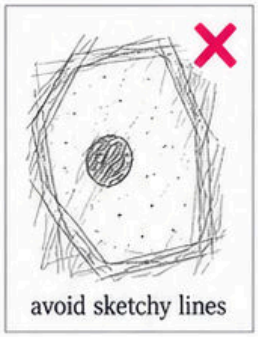
1 OBSERVED ONION-CELL CLUSTER

Draw the visible outline of several cells and only structures you can identify. Use a sharp pencil; maintain observed proportions. A micrograph is evidence: do not add every structure from a textbook.



Onion epidermal cells

Magnification: ×400



avoid sketchy lines

2 SCHOOL DRAWING CHECKLIST

11 marks total.

1	Drawing occupies at least half the page	[1]
2	Clear single lines, no overlapping sketch lines	[1]
3	No shading	[1]
4	Pencil label lines that do not overlap	[1]
5	All drawn parts labelled	[1]
6	Clear, underlined title	[1]
7	Magnification shown	[1]
8	Calculation of field-of-view width	[2]
9	Calculation of one cell's length	[2]

3 KEY IDEAS

Magnification: how many times larger the image is than the real object.

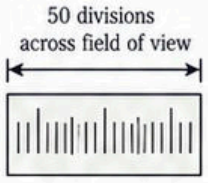
Resolution: ability to distinguish two close points as separate. More magnification alone does not guarantee more detail.

Light vs electron: Electron microscopes have greater resolving power and higher maximum useful magnification, so finer subcellular detail can be studied. Standard school light microscopes cannot resolve ribosomes. Mitochondria are difficult to distinguish in a routine unstained classroom preparation; visibility depends on resolution, staining and specimen preparation.

4 EXAMPLE CALCULATIONS

a Field-of-view width

A stage micrometer shows that 50 divisions (each 10 µm) span the width of the field of view at ×400 magnification. Calculate the field-of-view width.



Field-of-view width = number of divisions × size of one division

= 50 × 10 µm

= 500 µm = 0.50 mm

b Length of one cell

A cell measures 6 mm on the drawing. The magnification is ×400. Calculate the real length of the cell.

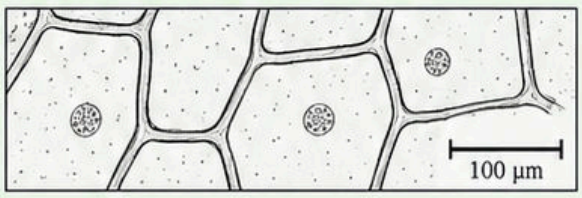
Actual size = image size ÷ magnification

= 6 mm ÷ 400

= 0.015 mm = 15 µm

5 SCALE BAR TIP

A scale bar resized together with its image stays valid. A printed magnification value can become wrong after resizing.



6 GCSE-STYLE QUESTION (2 marks)

Explain why an electron microscope reveals more cell detail.

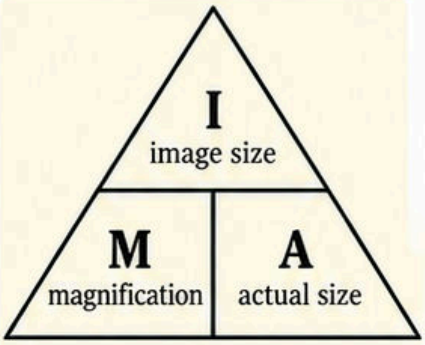
MODEL ANSWER

It has greater resolving power [1], so structures very close together can be distinguished as separate [1].

BIOLOGY REVISION NOTES

MAGNIFICATION AND UNIT CONVERSION

1 KEY FORMULAS



$$M = \frac{I}{A}$$

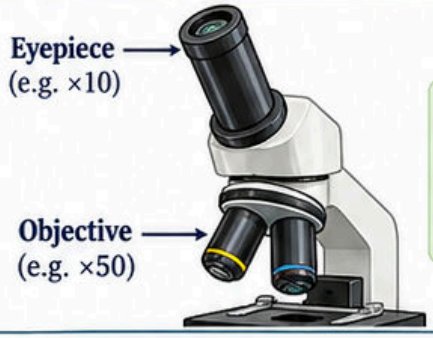
$$A = \frac{I}{M}$$

$$I = M \times A$$

I = image size
A = actual size
M = magnification

2 TOTAL MAGNIFICATION

Total microscope magnification = eyepiece magnification × objective magnification.



Magnification has no unit; write x400, not 400 micrometres.

3 UNIT CONVERSION

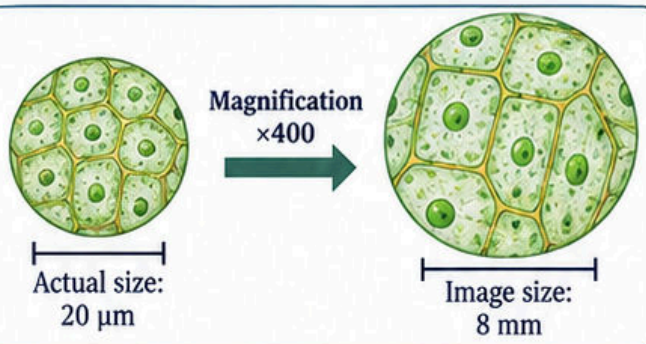
1 mm = 1000 micrometres (µm).
1 µm = 1000 nanometres (nm).

- mm → µm: multiply by 1000.
- µm → mm: divide by 1000.

MEMORY

Smaller unit, bigger number.

Convert image and actual size into the same unit before dividing.



4 WORKED EXAMPLES

Example 1 – Total magnification (2 marks)

A microscope has an eyepiece magnification of ×10 and an objective magnification of ×50. Calculate the total magnification.



Total magnification = 10 × 50 [1]
= ×500 [1]

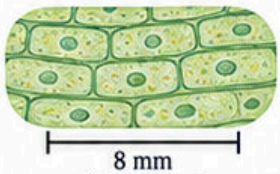
Example 2 – Find magnification (3 marks)

An image is 8 mm wide. The actual cell width is 20 µm. Calculate the magnification.

8 mm = 8000 µm [1]

$$M = \frac{8000}{20}$$ [1]

= ×400 [1]



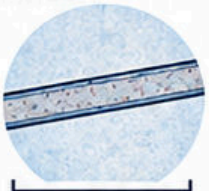
Example 3 – Find actual size (3 marks)

A capillary image is 5 mm across at a magnification of ×1000. Find the actual width in µm.

$$A = \frac{5}{1000} = 0.005 \text{ mm}$$ [1]

0.005 × 1000 [1]

= 5 µm [1]

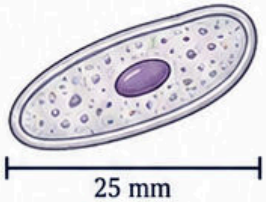


Example 4 – Draw an image (2 marks)

A real cell is 0.01 mm long. Calculate the length of the drawing at ×2500.

$I = 2500 \times 0.01$ [1]

= 25 mm [1]



5 HOW TO SECURE METHOD MARKS

1. Write the correct formula.
2. Substitute the values.
3. Show any unit conversion.
4. Keep extra digits until the final step.
5. Give the final answer with the correct unit.

6 COMMON TRAP

19.5 mm / 1500 = 0.013 mm = 13 µm.

The school review slide gives 7 µm for this expression; that arithmetic is incorrect.

7 QUICK CHECK

An image is 12 mm. The actual cell size is 30 µm. What is the magnification?

12000 / 30 = ×400.

BIOLOGY REVISION NOTES

FIELD OF VIEW AND CELL SIZE.

1 FIELD OF VIEW AND MAGNIFICATION

KEY IDEA: Increasing magnification decreases the field-of-view diameter.

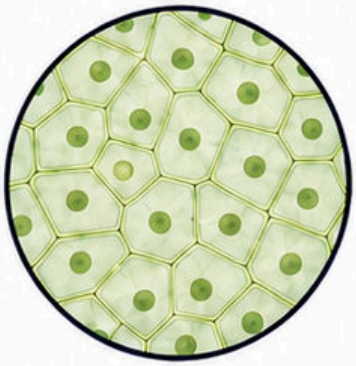
For the same microscope setup:

$$FOV_1 \times M_1 = FOV_2 \times M_2.$$

$$FOV_2 = \frac{FOV_1 \times M_1}{M_2}.$$

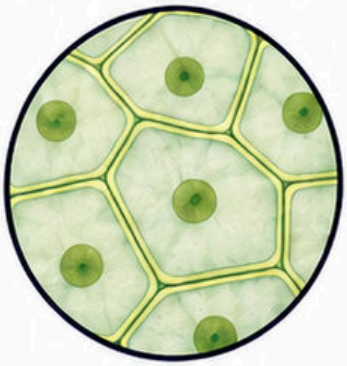
- Use total magnification consistently.
- Objective ratios also work if the eyepiece is unchanged.
- Increasing magnification means a smaller area is seen, so the field of view becomes smaller.

Low power (e.g. x100)



Larger field of view *(more cells visible).*

High power (e.g. x400)



Smaller field of view *(fewer, larger cells visible).*

2 ESTIMATE CELL SIZE

The actual cell length is approximately equal to the field-of-view diameter divided by the number of cell lengths spanning the diameter.

- Count along one line across the diameter, not every cell in the circular area.
- Include partial cells at the edges sensibly.
- Unequal cells make this an estimate.

3 WORKED EXAMPLE (4 marks)

The field of view is 1.6 mm at x100.
At x400, 10 cell lengths fit across the diameter.
Estimate one cell's length in µm.

Answer:

$$\text{FOV at x400} = \frac{1.6 \times 100}{400} \quad [1] \quad = 0.4 \text{ mm } [1].$$

$$\text{Cell length} = \frac{0.4}{10} \quad [1] \quad = 0.04 \text{ mm}.$$

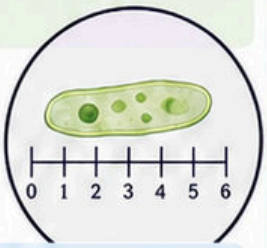
$$0.04 \times 1000 = 40 \text{ µm } [1].$$

4 MICROMETER EXAMPLE (2 marks)

Each calibrated division represents 0.01 mm.
A cell spans 5 divisions.

Answer:

Length = $5 \times 0.01 = 0.05$ mm [1]
= 50 µm [1].



Calibration depends on the objective: do not assume one eyepiece division always represents the same real length.

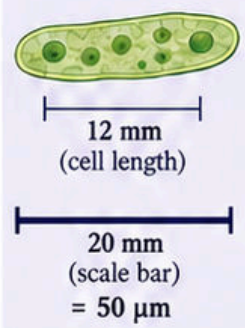
5 SCALE BAR EXAMPLE

A printed scale bar is 20 mm long and represents 50 µm.
A cell on the same image is 12 mm long.

Answer:

$$\text{Actual cell size} = \frac{12}{20} \times 50 = 30 \text{ µm}.$$

Both printed measurements must use the same unit.



6 STANDARD FORM

40 µm = 0.000040 m
= 4.0×10^{-5} m.

7 GCSE-STYLE TRY (3 marks)

The field of view is 2.0 mm at x100.
At x500, 8 cell lengths span the diameter.

Answer:

$$\frac{2.0 \times 100}{0.40} = 0.40 \text{ mm } [1];$$

$$\frac{0.40}{8} = 0.050 \text{ mm } [1] = 50 \text{ µm } [1].$$

BIOLOGY REVISION NOTES

GCSE-STYLE MINI TEST | TRY BEFORE PAGE 9

Original questions written for these school topics; not an official AQA paper.
Suggested time: 20 minutes. Total: 20 marks.
Before starting: cover the next page. Underline command words and circle units.
Write answers on separate paper.

1 Red blood cells and oxygen transport [4 marks]

Explain how two adaptations help a red blood cell transport oxygen.



2 Comparing xylem and phloem [4 marks]

Compare xylem and phloem in terms of: substance transported; direction of transport; whether conducting cells are living; one structural adaptation. Write paired comparisons.

Feature	Xylem	Phloem
Substance transported		
Direction of transport		
Are conducting cells living?		
One structural adaptation		

3 Preparing an onion-cell slide [4 marks]

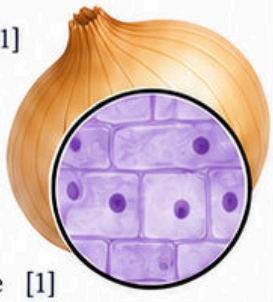
A student prepares an onion-cell slide and immediately selects the highest-power objective. Explain why each of these would improve the method:

(a) use a thin single layer of epidermis [1]

(b) add iodine solution [1]

(c) lower the coverslip at an angle [1]

(d) start with the lowest-power objective [1]



4 Calculating magnification [3 marks]

A cell image is 18 mm long. The actual cell is 45 µm long. Calculate the magnification. Show your working.

5 Using field of view to estimate cell length [3 marks]

A microscope's field-of-view diameter is 1.8 mm at x100. At x300, six cell lengths span the diameter. Estimate one cell's length in µm.

6 Resolution of a light microscope [2 marks]

Explain why increasing the magnification of a blurry light-microscope image may not reveal two nearby structures as separate.

State: give facts. **Explain:** link feature to reason. **Compare:** refer to both. **Calculate:** show working and units.

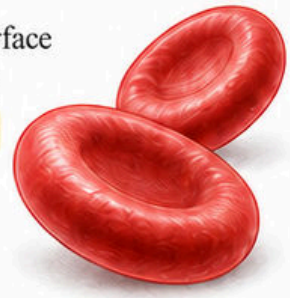
BIOLOGY REVISION NOTES

FULL-MARK ANSWERS | MINI TEST

Suggested marking points for our original questions.
These are not an official AQA mark scheme.

1 Q1 | 4 MARKS

Its **biconcave shape** gives a large surface area relative to volume [1], allowing rapid oxygen diffusion [1]. It has **no nucleus** [1], leaving more space for haemoglobin to bind and carry oxygen [1].



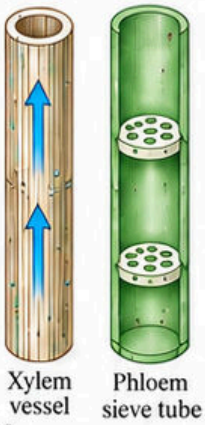
Why this scores: two named adaptations, each linked to an oxygen-transport benefit. 'More room for oxygen' is less precise than 'more room for haemoglobin'.

2 Q2 | 4 MARKS

Xylem transports water and mineral ions; phloem transports **dissolved sugars** [1].

Xylem transport is upwards from the roots; phloem transports from source to sink, upwards or downwards in different tubes [1].

Mature xylem vessel elements are dead; phloem sieve tube elements are living [1].



Xylem has hollow **lignified vessels**; phloem has perforated **sieve plates** that allow sugar solution to pass [1].

Why this scores: every point compares both tissues. Do not write 'phloem carries food' without naming dissolved sugars.

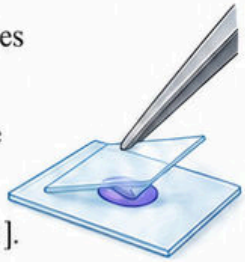
3 Q3 | 4 MARKS

(a) A **thin layer** lets light pass through and reduces overlapping cells [1].

(b) **Iodine** improves contrast so cell structures are easier to distinguish [1].

(c) **Angled lowering** reduces trapped air bubbles [1].

(d) **Low power** gives a larger field of view, making cells easier to locate [1].



4 Q4 | 3 MARKS

18 mm = **18000 µm** [1].

M = 18000 / 45 [1] = **x400** [1].

Magnification (M) = $\frac{\text{image size}}{\text{actual size}}$

5 Q5 | 3 MARKS

New FOV = 1.8 x 100 / 300 = **0.6 mm** [1].

Cell length = 0.6 / 6 = 0.1 mm [1] = **100 µm** [1].

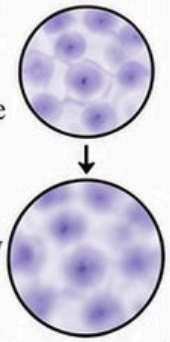


New field of view = $\frac{\text{original FOV} \times \text{original magnification}}{\text{new magnification}}$

6 Q6 | 2 MARKS

The **resolving power** of the microscope is limited [1].

Enlarging the image cannot separate details that were not resolved, so it may only enlarge the blur [1].



Enlargement cannot add new detail.

REVISION ROUTINE

Read → cover → recall aloud → attempt → mark → correct

Revisit missed points after 2 days and again after 7 days. Keep separate lists for missing knowledge, incomplete explanations and calculation errors.

Sources: school specialised-cell posters and RP1 slides supplied 17 October; AQA 8461 Cell biology specification. Roadmap-only later topics excluded.